# Cheatsheet AZ-104 — 15 · Azure Monitor: metrics, logs and KQL, alerts, Insights and Network Watcher

## 1. Azure Monitor map

| Data | What it is | Where it comes from | Where to query it |
|---|---|---|---|
| **Platform metrics** | Numeric series (CPU %, Disk IOPS, transactions...) every **1 min**, retained **93 days**, nothing to configure | All resources | **Metrics explorer** (avg / min / max / sum / count aggregation, split by dimension, pin to dashboard), metric alerts |
| **Guest OS metrics** | Memory, logical disk, processes inside the VM | **Azure Monitor Agent** + Data Collection Rule (or Diagnostics extension to storage) | Metrics explorer, Log Analytics (`InsightsMetrics`, `Perf`) |
| **Activity log** | Control plane: who created, deleted, modified, started; Service Health, Policy, Autoscale, Security | Automatic per subscription, **90 days** | Activity log blade; to Log Analytics (`AzureActivity`) with a diagnostic setting to retain longer |
| **Resource logs** | Data plane of the resource (StorageBlobLogs, AppServiceHTTPLogs, NSG, Key Vault audit...) | **Diagnostic settings** per resource | Log Analytics, Storage, Event Hub |
| **OS logs** | Event logs, Syslog, counters, files | AMA + DCR | `Event`, `Syslog`, `Perf` |
| **Application Insights** | App telemetry: requests, dependencies, exceptions, **availability tests**, Live Metrics | SDK / auto-instrumentation | `AppRequests`, `AppExceptions`...; Application Map |
| **Insights** | Ready-made views: **VM insights** (Performance + Map; AMA + Dependency agent), **Storage insights**, **Network insights**, Container insights | Enabled per resource | Monitor → Insights |

## 2. Diagnostic settings

| Destination | Purpose | Signal |
|---|---|---|
| **Log Analytics workspace** | Query with **KQL**, log alerts, workbooks, Insights | "query", "correlate", "alert on log" |
| **Storage account** | **Cheap archive** and long retention (days per category or lifecycle) | "retain for 7 years", "audit archive" |
| **Event Hub** | Streaming to a **SIEM** or third parties (Splunk, QRadar) | "third-party SIEM", "stream" |
| **Partner solution** | Datadog, Elastic... | — |

- Up to **5 diagnostic settings** per resource; each category only once per destination. Categories: logs (**allLogs** / **audit** groups or individual categories) and metrics. The **Activity log** is sent from Monitor → Activity log → Export (subscription-level diagnostic setting).
- VMs: the diagnostic setting only covers host metrics; anything **inside** the OS needs **AMA + DCR** (Log Analytics agent / MMA retired in 2024). The **DCR** defines what to collect (counters, event logs, Syslog, custom text / JSON, IIS) and to which workspace; it is associated with VMs, VMSS or Arc; needs a **managed identity** on the VM.

## 3. Log Analytics and KQL

| Aspect | Detail |
|---|---|
| **Workspace** | Own region, collects data from **any subscription and region**; priced per GB ingested (PAYG or **commitment tiers**) and per retention; **daily cap** to limit cost |
| **Retention** | Default **30 days** (90 for `AzureActivity`, `Usage` and App Insights tables); analytics up to **730 days**; **total retention** (long-term, queryable with search jobs) up to **12 years**; per workspace or **per table** |
| **Table plans** | **Analytics** (everything), **Basic** (cheap, 30 days, limited queries), **Auxiliary** (very cheap, high-volume logs) |
| **Access** | **Log Analytics Reader / Contributor** roles; *workspace-context* mode (the whole workspace) or *resource-context* (only data of the resources you have access to) |
| Tables that show up | `Heartbeat`, `Perf`, `Event`, `Syslog`, `InsightsMetrics`, `AzureActivity`, `AzureDiagnostics`, `AzureMetrics`, `SecurityEvent`, `StorageBlobLogs`, `AppRequests`, `Update` |

```
Heartbeat | where TimeGenerated > ago(1h) | summarize LastBeat = max(TimeGenerated) by Computer   // reporting VMs
Perf | where ObjectName == "Processor" and CounterName == "% Processor Time"
     | summarize avg(CounterValue) by Computer, bin(TimeGenerated, 5m) | render timechart
Event | where EventLevelName == "Error" and TimeGenerated > ago(24h) | summarize count() by Computer, Source | top 10 by count_
AzureActivity | where OperationNameValue has "delete" and ActivityStatusValue == "Success"
              | project TimeGenerated, Caller, ResourceGroup, OperationNameValue
InsightsMetrics | where Name == "AvailableMB" | summarize min(Val) by Computer                    // free memory (VM insights)
StorageBlobLogs | where StatusCode >= 400 | summarize count() by StatusText, bin(TimeGenerated, 1h)
search "vm-sql01" | where TimeGenerated > ago(1d) | distinct $table                              // which tables it appears in
```

- Operators: `where` (filters), `project` (columns), `extend` (computes), `summarize ... by` (groups; `count()`, `avg()`, `max()`, `dcount()`), `bin()` (time buckets), `sort by` / `order by`, `top N by`, `take` / `limit`, `distinct`, `join`, `parse`, `render`, `let`. Time: `ago(1h)`, `between`, `startofday()`. Text comparison: == exact, =~ case-insensitive, `has` (whole word, fast), `contains` (substring), `startswith`. Save: **queries**, **functions**, **workbooks**.

## 4. Alerts

| Rule type | Signal | How it evaluates | When |
|---|---|---|---|
| **Metric** | Platform, guest or custom metric | Every **1 min**; aggregation window (e.g. 5 min), **static** or **dynamic** (ML) threshold; one rule for **several resources** of the same type and region; **stateful** (resolves after 3 OK evaluations) | "CPU > 80 % for 5 minutes", near real-time |
| **Log search** | **KQL** query on the workspace | Frequency 1 min to 1 day; condition by **number of results** or **measure** (metric from the query); **split by dimensions** (one alert per resource); Basic / Auxiliary tables are not supported | "event ID 4625 more than 5 times", "service stopped" |
| **Activity log** | Control operations (Administrative), **Service Health**, **Resource Health**, Policy, Autoscale, Security | **Stateless**: every event fires; filter by operation and status | "when a VM is deleted", "planned maintenance in West Europe", "resource became unavailable" |
| **Smart detection / Prometheus** | App Insights anomalies / Prometheus metrics | — | Apps / AKS |

| Piece | Detail |
|---|---|
| **Rule** | Scope (resource, RG, subscription) + signal and condition + **severity 0 (critical) - 4 (verbose)** + actions + name; **Monitoring Contributor** (create) / **Monitoring Reader** (view) roles; fired alerts are kept **30 days** |
| **Action group** | Reusable, **global**, up to **5 per rule**: **email**, **SMS**, **push** (Azure app), **voice** notifications, **Email Azure Resource Manager role** (Owner, Contributor, Reader, Monitoring Contributor / Reader of the subscription); **Automation runbook**, **Azure Function**, **Logic App**, **Event Hub**, **ITSM**, **webhook** / **secure webhook** (Entra ID) actions; **common alert schema**; rate limiting on SMS / email / voice |
| **Alert processing rules** | Applied to already fired alerts within a scope: **suppress notifications** (maintenance window, by schedule) or **add action groups** to a whole RG / subscription without touching each rule |
| **States** | *User response*: New → Acknowledged → Closed · *Monitor condition*: Fired / Resolved (stateful) |

- "Notify the team when the VM is deleted" → activity log alert (Administrative, Delete Virtual Machine, Succeeded). "Get notified of Azure outages in my region" → **Service Health alert** with action group. "Run a runbook when CPU is high" → metric alert + action group with Automation runbook. "Stop emails during the maintenance window" → alert processing rule. "Alert on a KQL result" → log search alert. "The VM alert must also apply to new VMs in the RG" → metric alert scoped to the **resource group** (multi-resource).

## 5. Insights and Network Watcher

| Tool | What it does | Requirements / detail |
|---|---|---|
| **VM insights** | Performance (CPU, memory, disk, network) and **Map** (processes and connections) | AMA + **Dependency agent**; `InsightsMetrics` table; a dedicated DCR |
| **Storage insights** | Transactions, latency, availability, capacity per account; errors | Free metrics; logs with diagnostic settings (`StorageBlobLogs`...) |
| **Network insights** | Topology, health of LB / App Gateway / gateways, **Connection monitor**, Traffic analytics | Network Watcher |
| **Connection monitor** | **Continuous** tests from VMs (Network Watcher extension) or Arc to a VM, IP, FQDN or URL over TCP / ICMP / HTTP; **% checks failed** and **round-trip time** metrics; alerts | Replaces Network performance monitor |
| **IP flow verify** / **NSG diagnostics** / **Next hop** / **Effective security rules** | One-off diagnosis of NSGs and routes | Cheatsheet 12 |
| **Connection troubleshoot** | One-off source → destination test with hops and latency | Network Watcher extension on the source |
| **Packet capture** | pcap capture on a VM / VMSS with filters and limit (5 h), to storage or disk | Extension |
| **VPN troubleshoot** | Diagnosis of gateways and connections | Logs to storage |
| **Flow logs** + **Traffic analytics** | **VNet** flow logging (NSG flow logs are retired in Sept. 2027) and analysis in Log Analytics | Storage account + workspace |
| Limits | **1 Network Watcher per region and subscription** (RG `NetworkWatcherRG`), enabled automatically when a VNet is created; 100 connection monitors per region | — |

## 6. Commands that show up

| Task | PowerShell (Az.Monitor / Az.OperationalInsights / Az.Network) | Azure CLI |
|---|---|---|
| Workspace | `New-AzOperationalInsightsWorkspace -ResourceGroupName RG1 -Name law1 -Location westeurope -Sku PerGB2018 -RetentionInDays 90` · `Set-AzOperationalInsightsWorkspace -RetentionInDays 365` · `Update-AzOperationalInsightsTable -TableName Event -RetentionInDays 30 -TotalRetentionInDays 730` | `az monitor log-analytics workspace create -g RG1 -n law1 --retention-time 90` · `az monitor log-analytics workspace update --retention-time 365` · `az monitor log-analytics workspace table update -n Event --retention-time 30 --total-retention-time 730` |
| Query | `Invoke-AzOperationalInsightsQuery -WorkspaceId <id> -Query "<KQL query>"` | `az monitor log-analytics query -w <id> --analytics-query "<KQL query>"` |

| Task | PowerShell (Az.Monitor / Az.OperationalInsights / Az.Network) | Azure CLI |
|---|---|---|
| Diagnostic setting | `New-AzDiagnosticSetting -Name ds1 -ResourceId <id> -WorkspaceId <law id> -Log (New-AzDiagnosticSettingLogSettingsObject -CategoryGroup allLogs -Enabled $true) -Metric (New-AzDiagnosticSettingMetricSettingsObject -Category AllMetrics -Enabled $true)` | `az monitor diagnostic-settings create -n ds1 --resource <id> --workspace <law id> --logs '[{"categoryGroup":"allLogs","enabled":true}]' --metrics '[{"category":"AllMetrics","enabled":true}]'` · to storage: `--storage-account st1` · to Event Hub: `--event-hub eh1 --event-hub-rule <auth rule id>` |
| Activity log | `Get-AzActivityLog -ResourceGroupName RG1 -StartTime (Get-Date).AddDays(-1)` (formerly `Get-AzLog`) | `az monitor activity-log list -g RG1 --offset 1d --query "[].{op:operationName.value, who:caller}"` |
| Action group | `$email = New-AzActionGroupEmailReceiverObject -Name ops -EmailAddress ops@contoso.com; New-AzActionGroup -ResourceGroupName RG1 -Name ag1 -ShortName ag1 -Location global -EmailReceiver $email` | `az monitor action-group create -g RG1 -n ag1 --short-name ag1 --action email ops ops@contoso.com --action sms oncall 34 600000000` |
| Metric alert | `$cond = New-AzMetricAlertRuleV2Criteria -MetricName "Percentage CPU" -TimeAggregation Average -Operator GreaterThan -Threshold 80; Add-AzMetricAlertRuleV2 -Name cpu80 -ResourceGroupName RG1 -TargetResourceId <vm id> -Condition $cond -WindowSize 00:05:00 -Frequency 00:01:00 -Severity 2 -ActionGroupId <ag id>` | `az monitor metrics alert create -n cpu80 -g RG1 --scopes <vm id> --condition "avg Percentage CPU > 80" --window-size 5m --evaluation-frequency 1m --severity 2 --action <ag id>` |
| Log alert | `New-AzScheduledQueryRule -Name errs -ResourceGroupName RG1 -Location westeurope -Scope <law id> -CriterionAllOf (New-AzScheduledQueryRuleConditionObject -Query '<KQL query>' -TimeAggregation Count -Operator GreaterThan -Threshold 5) -WindowSize 00:15:00 -EvaluationFrequency 00:05:00 -Severity 2 -ActionGroupResourceId <ag id>` | `az monitor scheduled-query create -n errs -g RG1 --scopes <law id> --condition "count 'Placeholder_1' > 5" --condition-query Placeholder_1="<KQL query>" --window-size 15m --evaluation-frequency 5m --severity 2 --action-groups <ag id>` |
| Activity log alert | `New-AzActivityLogAlert -Name vmdel -ResourceGroupName RG1 -Scope /subscriptions/<id> -Condition (New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -Equal Administrative -Field category),(... -Equal Microsoft.Compute/virtualMachines/delete -Field operationName) -Action (New-AzActivityLogAlertActionGroupObject -Id <ag id>)` | `az monitor activity-log alert create -n vmdel -g RG1 --scope /subscriptions/<id> --condition category=Administrative and operationName=Microsoft.Compute/virtualMachines/delete --action-group <ag id>` |
| AMA and DCR | `Set-AzVMExtension -Publisher Microsoft.Azure.Monitor -ExtensionType AzureMonitorWindowsAgent -Name AMA -EnableAutomaticUpgrade $true ...` · `New-AzDataCollectionRule` · `New-AzDataCollectionRuleAssociation -TargetResourceId <vm id> -DataCollectionRuleId <dcr id> -AssociationName a1` | `az vm extension set -n AzureMonitorWindowsAgent --publisher Microsoft.Azure.Monitor --vm-name vm1 -g RG1 --enable-auto-upgrade true` · `az monitor data-collection rule create ... --rule-file dcr.json` · `az monitor data-collection rule association create -n a1 --rule-id <dcr id> --resource <vm id>` |
| Network Watcher | `Get-AzNetworkWatcher` · `New-AzNetworkWatcherConnectionMonitor` · `New-AzNetworkWatcherPacketCapture -NetworkWatcher $nw -TargetVirtualMachineId <id> -PacketCaptureName pc1 -StorageAccountId <id> -TimeLimitInSeconds 300` · `Test-AzNetworkWatcherIPFlow` · `Get-AzNetworkWatcherNextHop` | `az network watcher configure -l westeurope --enabled true` · `az network watcher connection-monitor create` · `az network watcher packet-capture create -g RG1 --vm vm1 -n pc1 --storage-account st1 --time-limit 300` · `az network watcher test-connectivity --source-resource vm1 --dest-address www.contoso.com --dest-port 443` · `az network watcher flow-log create --vnet vnet1 --storage-account st1 --workspace law1 --traffic-analytics true` |

## 7. Decision tree

```
What does the wording ask for?
├─ See CPU / IOPS of a resource right now ..... Metrics explorer (platform metrics)
├─ Free memory or logical disk of a VM ..... AMA + DCR (guest metrics) or VM insights
├─ Who deleted a resource ..... Activity log (90 days) → to Log Analytics for more
├─ Retain audit logs cheaply for years ..... Diagnostic setting → Storage account
├─ Send logs to an external SIEM ..... Diagnostic setting → Event Hub
├─ Query and correlate with KQL ..... Diagnostic setting → Log Analytics
├─ 2 years queryable / 7 years total retention ..... Workspace or table: analytics 730 d + total retention
├─ Alert "CPU > 80 % 5 min" on all VMs in the RG ..... Metric alert scoped to the RG
├─ Alert on text in the event log ..... Log search alert (KQL)
├─ Alert when a resource is deleted / modified ..... Activity log alert (Administrative)
├─ Notify about Azure incidents in my region ..... Service Health alert
├─ Run a runbook / function when it fires ..... Action group (Automation runbook / Function)
├─ Silence alerts during the maintenance window ..... Alert processing rule (suppress)
├─ Process and dependency map of a VM ..... VM insights (Map)
├─ Watch VM → service latency continuously ..... Connection monitor
└─ Capture packets from a VM ..... Network Watcher packet capture
```

## 8. Signals → answer

| Wording | Answer |
|---|---|
| "metrics older than 93 days" | Send them to Log Analytics (diagnostic setting, AllMetrics) |
| "activity log for 1 year" | Activity log diagnostic setting to Log Analytics (retention) or Storage |
| "memory usage of the VM in Metrics" | Guest metrics: AMA + DCR (or VM insights) |
| "Log Analytics agent / MMA" | Retired: migrate to Azure Monitor Agent + DCR |
| "collect IIS logs / custom log file" | DCR with custom text logs |
| "query logs from VMs in several subscriptions" | One central workspace (collects from any subscription) |
| "users should see only logs of their resources" | Resource-context access (permission on the resource) |

| Wording | Answer |
|---|---|
| "reduce ingestion cost for verbose logs" | Basic / Auxiliary table, daily cap, commitment tier |
| "keep audit logs for 10 years" | Total retention (up to 12 years) or archive in Storage |
| "alert when VM CPU > 80 % for 10 minutes" | Metric alert, aggregation granularity 10 min, average > 80 |
| "alert that also applies to VMs created later" | Scope at RG / subscription (multi-resource) |
| "alert when Windows service stops" | Log search alert on `Event` (ID 7036) |
| "alert when a resource group is deleted" | Activity log alert, delete operation |
| "notify Owners of the subscription" | Email Azure Resource Manager role = Owner in the action group |
| "same action group for 50 alert rules without editing each" | Alert processing rule that adds the action group |
| "no alerts during the weekly maintenance" | Suppression alert processing rule with recurrence |
| "alert severity for a critical outage" | Sev 0 |
| "test connectivity continuously from VM to on-prem SQL" | Connection monitor (TCP 1433) |
| "capture traffic on the VM for support" | Packet capture (Network Watcher extension) |
| "Network Watcher not available in the region" | Enable it (one per region and subscription) |

## 9. Mistakes to avoid

1. Looking for memory or logical disk in a VM's platform metrics: an agent is needed.
2. Sending to Storage what later has to be queried with KQL (or to Log Analytics what only needs cheap archiving).
3. Sticking with the Log Analytics agent (MMA) or the diagnostics extension for OS logs.
4. Assuming the Activity log lasts more than 90 days without exporting it.
5. Creating one metric alert per VM when a single one scoped to the RG covers the new ones.
6. Using a log alert for "VM deleted" (that is Activity log) or a metric alert for event log text (that is log search).
7. Editing 50 rules to add an action group instead of an alert processing rule.
8. Forgetting the Network Watcher extension for packet capture / connection troubleshoot / Connection monitor.